

From theory to macroecological evidence: a global test of ISLAND SYNDROME in plant floral traits

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BACKGROUND · THE CLASSIC EXPECTATION

Isolation filters arrivals.
Plants may face unreliable pollination.
Island syndrome: recurring traits of island biotas.
Baker's law: reproduction without a mate aids establishment.
Expected flowers: more selfing, less showy, easier access.

GAP · GLOBAL PATTERNS AND MECHANISMS

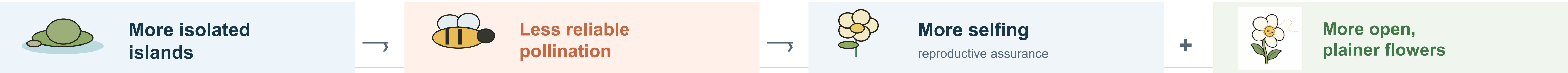
Does the floral island syndrome recur globally, and what processes can produce it?
Colour attracts visitors. Does its isolation response simply accompany selfing, or reflect another pathway?
Global colour–structure–breeding tests remain limited.
Q1: test the phenomenon. Q2: understand a possible process.

CONCLUSIONS · ANSWERS TO Q1 AND Q2

Q1 More isolated floras are more self-compatible. Colour and structure respond differently among regions.
Q2 Isolation changes investment in simulated islands. Small populations and visitor history modify the response.
Evidence: 4,379 observed islands; 128 simulated visitor histories.

Q1 Global evidence: isolation, floral traits and pollen limitation

CLASSIC HYPOTHESIS · Proposed links, not an established causal chain



1 Floral patterns (H1 · H2)

SELF-BUILT GLOBAL FLORAL DATABASE
106,295 angiosperm species
222,688 resolved species × axis entries
4,379 islands analysed in 4 regions
BUILD: exact island polygons → GBIF flora → trait evidence
Geography: GSHHO 2.3.7 [1]. Occurrences: GBIF [2].
Traits: floras (eFloras, BHL), published studies and trait datasets (Pladias, BIOLFLOR, AusTraits) [3–5].
Harmonize names and trait states; retain source provenance. Evidence tiers retained; direct-only sensitivity.
7 TRAITS: reproduction (3), colour (1), structure (3)
Self-compatibility, mating system, autonomous selfing; plain colour, generalized form, radial symmetry, open tube. 69.63% coverage. Missing traits remain missing.

H1 · METHOD
Y: trait-positive species out of scored species.
X: isolation D + controls C.
 $\text{logit}(p) = \alpha + \beta D + \gamma C$
Beta-binomial, maximum likelihood. Fit each of 7 traits in each region.
Spatial-block t tests (G–1 df).
 $\beta > 0$: trait more common with greater isolation.

D = standardized $\ln(1 + \text{coast distance in km})$, C = standardized $\log(\text{area})$, climate PC1–4. Spatial-block 95% t CIs. Dots = islands. No pooled syndrome test.
*, **, ***: nominal $p < .05$, .01, .001; ns ≥ .05.

H1 How do traits vary with isolation?

H2 · METHOD
Y: access score (linear model) or plain colour (logit).
X: D + selfing score S + C.
Test isolation after adjusting S.
S uses reproductive traits only.
FDR across 8 region × trait tests.
Access: $Y = \alpha + \beta D + \delta S + \gamma C$. Colour: $\text{logit}(p) = \alpha + \beta D + \delta S + \gamma C$.

STRUCTURE: floral access
N. mid-latitude: $q = 0.176$
N. high-latitude: $q = 0.002$
Tropical: $q = 0.019$
S. extratropical: $q = 0.298$

COLOUR: plainness
N. mid-latitude: $q = 0.978$
N. high-latitude: $q = 0.817$
Tropical: $q = 0.088$
S. extratropical: $q = 0.002$

Structure shifts remain in N. high-latitude/tropics: colour shifts in the south. Direct pollinator selection / filtering may also act beyond selfing. Filled: $q < .05$. Alternatives are plausible; direct selection remains untested.

N. mid-latitude

Red/pink declines. Within yellow/orange, bee-associated forms decline ($q < .001$).

POSSIBLE ECOLOGICAL EXPLANATION
Handling and tongue length filter visitors. Fewer bee-associated forms may favour flowers usable by remaining insects: functional turnover, not proven bee loss.

N. high-latitude

Within blue/purple flowers, tubular forms / deeper tubes decline ($q < .001$).

POSSIBLE ECOLOGICAL EXPLANATION
Shallow rewards are usable by short-tongued flies, effective Arctic pollinators. Open flowers may still rely on few partners. Butterflies are present at high latitudes.

H2 selfing-core score: positive in 4 regions (2 $p < .05$). South: generalized form ↑, shallow tubes ↓. Tropical Direct access: $q = .127$.

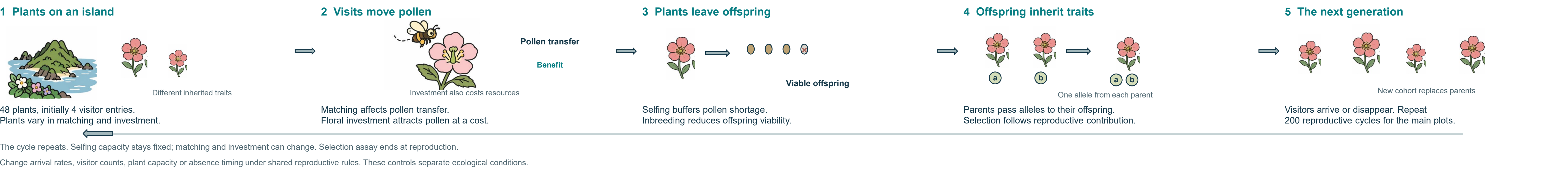
H4 atomic traits: autonomous selfing $p = 4.85 \times 10^{-8}$; radial symmetry $p = 1.47 \times 10^{-8}$; generalized form $p = .0454$; self-compatibility $p = .150$.

PATTERN TO PROCESS Q1 establishes isolation-associated shifts across floras. Q2 tests a possible mechanism after plants establish. In Q2, investment can change with fixed selfing capacity. Flower colour and the four regional visitor guilds are not explicitly simulated.

Q2 How does isolation become a floral response?

Q2 · Simulated islands: how do isolation and history change floral evolution?

ONE MODEL. Floral change follows pollen transfer, reproductive trade-offs and inheritance; no direction of change is imposed.



GENERAL PRINCIPLE · Isolation changes the benefit of floral investment Isolation limits visitor arrivals. Small populations show less floral divergence than the deterministic reference.

SELECTION DIAGNOSIS + TWO PARALLEL EVOLUTION MODELS

Selection assay: slightly change investment in a fixed population. Does expected maternal + paternal contribution rise? No generations advance.

Evolution: both restart from the same founders and visitor history.

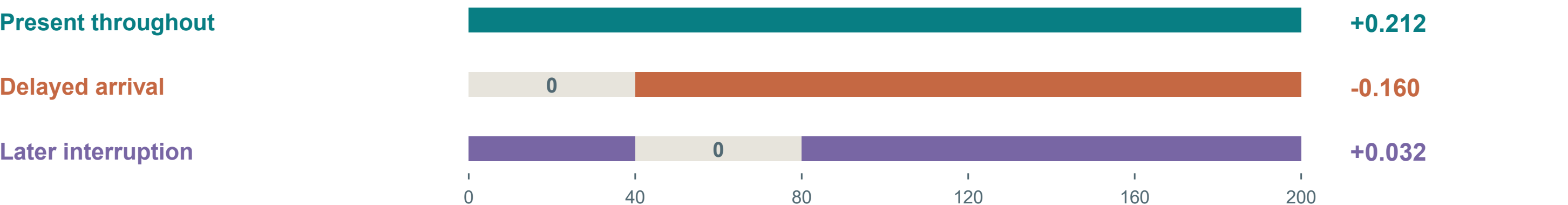
DENSITY · EXPECTED EVOLUTION
Expected counts of genotype groups. Reproduction and inheritance without random birth / recruitment draws.
A: 2.4 B: 0.6 (expected counts)
Examples are schematic. Density also differs in finite pollen self-exclusion; it is not the exact ABM mean.

ABM · REALIZED EVOLUTION
One agent = one individual plant. Sample offspring, inherited alleles and recruitment; repeats can differ.
AAB or AAA: a variant may be absent

WHY THESE CONTROLS?
R1: arrivals decline with isolation. R2: match counts to test that route.
R3: raise plant capacity (48 → 192) to test finite-population effects.
READ THE PLOTS: near – far investment change; >0 = far lower.
Orange = density; teal = ABM. Horizontal grey = no isolation effect.

UNEXPECTED RESULT · Returning visitors do not erase floral history within 200 cycles

H · Does the timing of visitor absence leave a floral legacy?



4 visitor entries when present, 0 in gaps. Last 120 cycles shared. Gap arms differ in time since return: 160 vs 120 cycles.

Longer check: 400-cycle initial gap + 1,600 shared cycles; gap – uninterrupted = -0.462 ABM / -0.004 density ($\mu = 10^{-4}$). Finite duration, not proof of irreversibility.

DISCUSSION · What makes an island syndrome?

Shared filtering, diverse floral solutions

Isolation is associated with greater pollen limitation and more self-compatible floras. Selfing and accessible flowers are associated with lower pollen limitation (H4).

Q2 · Isolation changes the return on floral investment

With selfing capacity held fixed, visitor conditions still alter investment evolution. Thus, reduced investment need not require increased selfing capacity in the model.

Q2 · Selection does not guarantee recovery

Small populations evolve less than density predictions; visitor history leaves a legacy. Island differences may reflect population size and history as well as current isolation.

SI · MODEL DETAILS $A_{w,t} = (0.1 + z) \exp[-((x - \theta_w)/b_w)^2]$; $O_t = B \exp(-c_z z^2 - c_a a^2)$; $F_t = O_t[1 - \exp(-R_t/(2P_{d,t}))]$; $S_t = a_t(O_t - F_t)(1 - \delta)$.

x: access; z: investment; θ_w : visitor optimum/breadth; A: affinity; O: ovules; R: received pollen; F: viable selfed seeds; S: viable selfed seeds; a: selfing capacity; δ : depression. Delayed-selfing equations.

Composition: 5 access states × 5 pools (4 / 8 visitor entries), 12 repeats, 60 cycles. Isolation: 200 cycles; 128 histories × 3 starts × 4 interventions × 2 distances = 24,576 pairs. $a = \delta = c_z = 0.5$; $K = 48 / 192$.

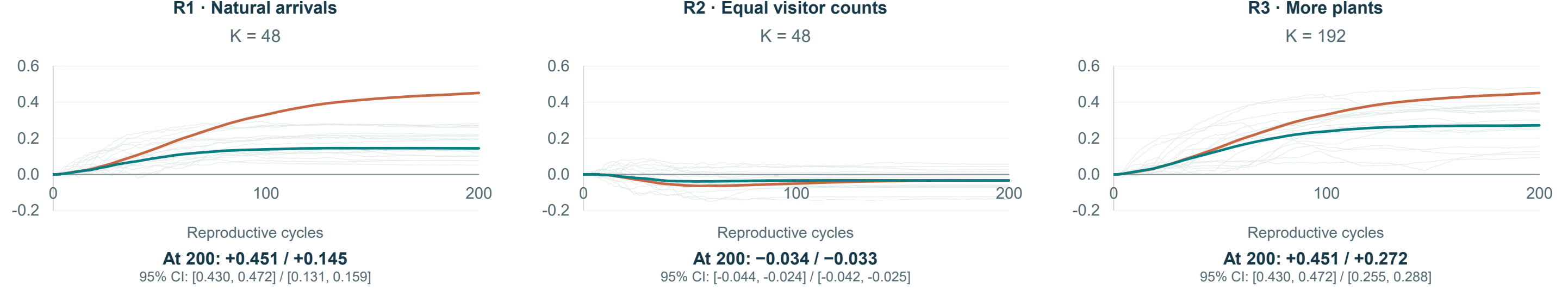
DATA SOURCES [1] Weasel & Smith 1996, GSHHG; [2] GBIF.org; [3] eFloras; BHL; [4] Chytrý et al. 2021, Pladias; Klotz et al. 2002, BIOLFLOR; [5] Falster et al. 2021, AusTraits; [6] Bennett et al. 2018, GLoPL.

SIMULATION OUTPUT · R1–R3

200 cycles, 128 independent visitor histories; 3 starts × 8 ABM repeats within each history.

Density: deterministic prediction

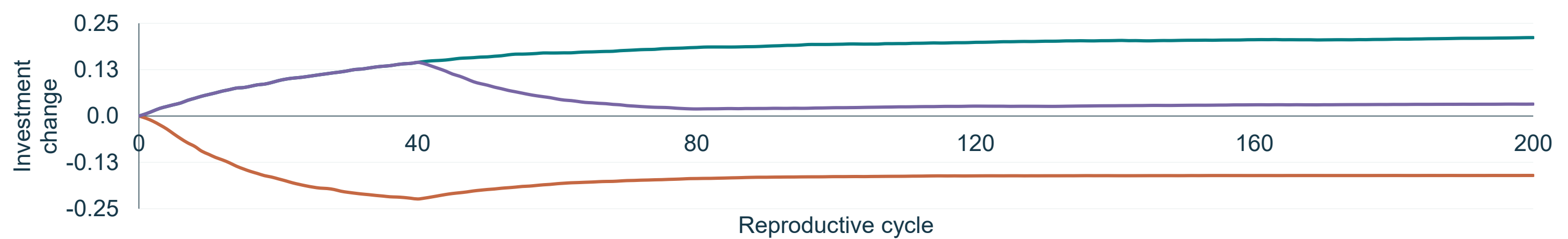
ABM: sampled individual plants



R1: fewer arrivals to far islands produce a positive near – far investment difference.

R2: thinning to equal visitor counts nearly removes the difference; it also changes visitor persistence.

R3: more plants narrow the density–ABM gap. Sampling and finite mating remain distinct mechanisms.



READ H: colours match the timelines. Above / below 0 = investment increase / decrease. All populations survive. 128 pools × 2 ABM repeats. Both matching and investment inherit.

IZU · A nearby view of the same problem

Autonomous selfing

Oshima 44% (81165)
Hagiwara 37% (21060)
Miyake 37% (21060)
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Oshima 44% (81165)

Plain colour

Oshima 44% (81165)
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Hagiwara 37% (21060)
Oshima 44% (81165)

Shallow / open tube

Oshima 44% (81165)
Hagiwara 37% (21060)
Miyake 37% (21060)
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Q1: 7 islands extracted from our flora database; labels = positive / classified species.
Q2 field illustration: other islands – Oshima; matching lower in all 8 targets, but pollen responses split 4 lower / 4 higher; tube responses 3 shorter / 4 longer / 1 equal.
Separate contemporary field contrasts, not a test of evolution. Hiraiwa & Ushimaru 2024 (SI).